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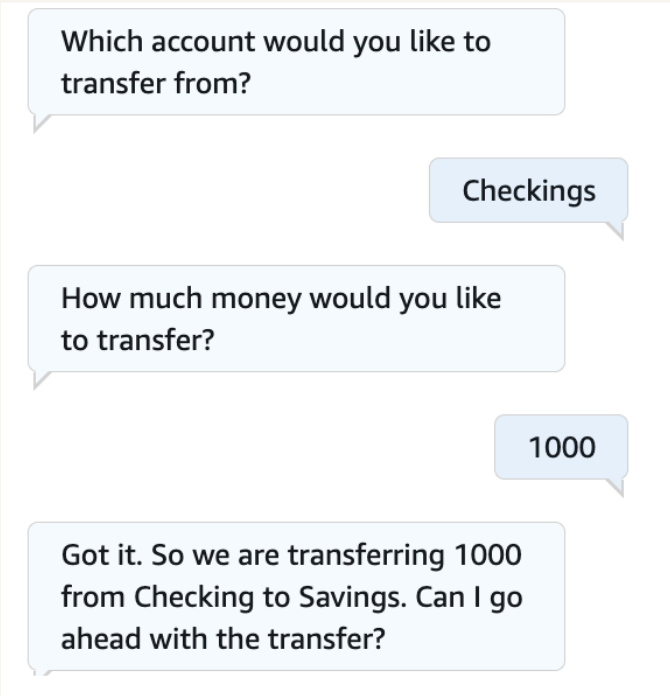
Build a Chatbot with Multiple Slots

CH Arpita Chowdhury



Introducing Today's Project!

What is Amazon Lex?

Amazon Lex is a service from Amazon Web Services that lets you build conversational chatbots using text and voice, and it is useful because it automatically understands user intent, manages conversation flow, and integrates seamlessly with services like AWS Lambda and CloudFormation to build real, production-ready chatbot solutions.

How I used Amazon Lex in this project

In today’s project, I used Amazon Lex to build and deploy a complete banking chatbot that supports balance checks, money transfers, context-aware conversations, and seamless intent transitions using the visual builder and CloudFormation.

One thing I didn't expect in this project was...

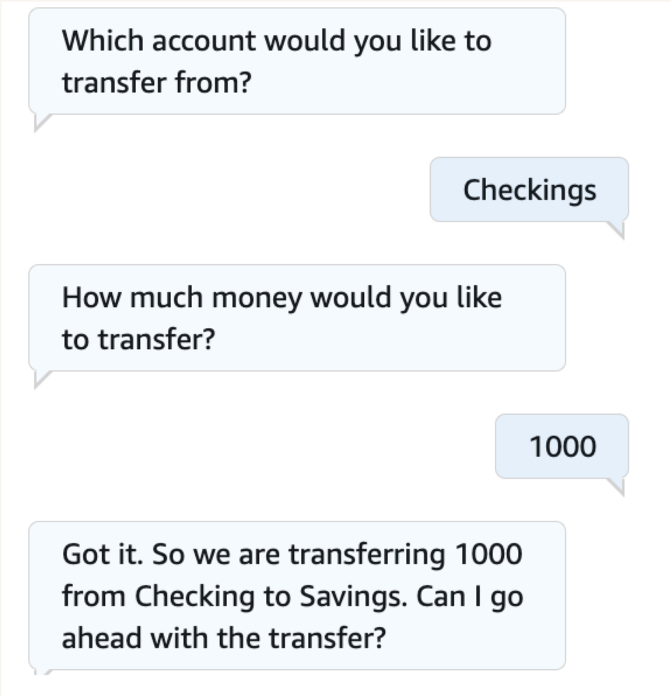
One thing I didn’t expect in this project was how powerful the visual builder and CloudFormation are together, and how quickly a complex, multi-intent chatbot can be deployed and updated with just a few configuration changes.

This project took me...

This project took me 2 hours.

TransferFunds

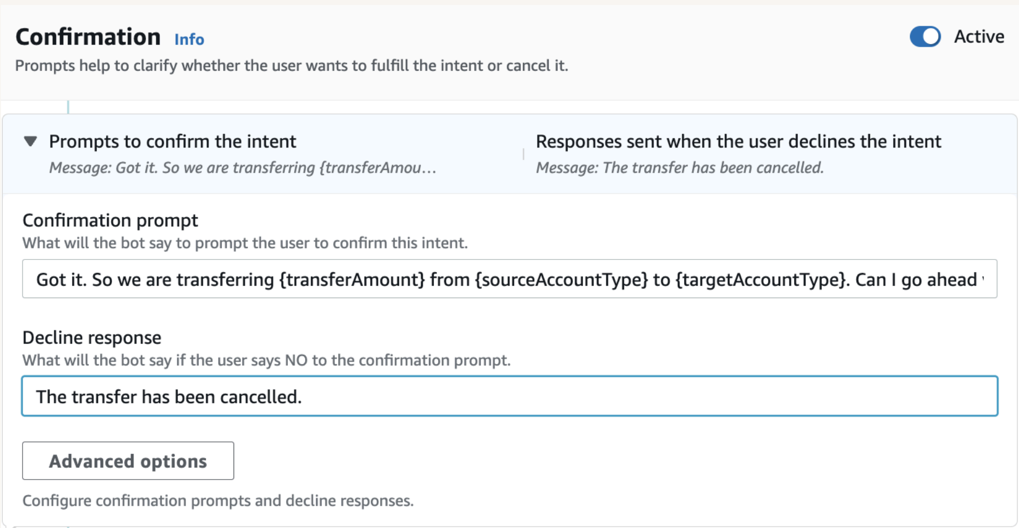
An intent I created for my chatbot was TransferFunds, which allows users to move money from one account to another by capturing two account types (source and destination) and the transfer amount in a single conversational flow.



Using multiple slots

For this intent, I had to use the same slot type twice because transferring money requires two different account types one to identify the source account the money is coming from and another to identify the target account the money is being sent to.

I also learnt how to create confirmation prompts, which are messages the chatbot uses to double-check a user's intent before completing an action, helping prevent mistakes by asking the user to confirm or cancel the request (such as approving or cancelling a money transfer).



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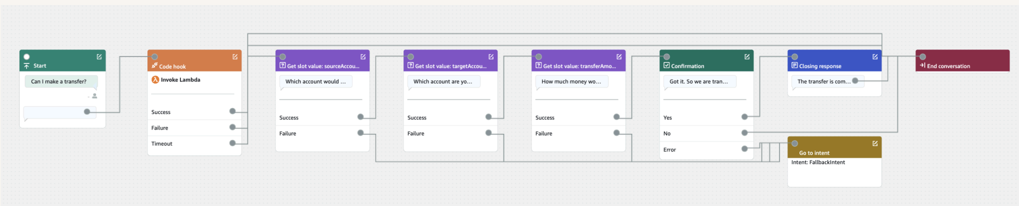
Exploring Lex features

Lex also has a special conversation flow feature that visually shows how users move through intents, slots, confirmations, and responses, making it easier to understand, design, and debug the chatbot's overall interaction logic.

You could also set up your intent using a visual builder! A visual builder is a drag-and-drop interface in Amazon Lex that lets you design and manage conversation flows visually, making it easier to see how intents, slots, and confirmations connect without writing everything manually.



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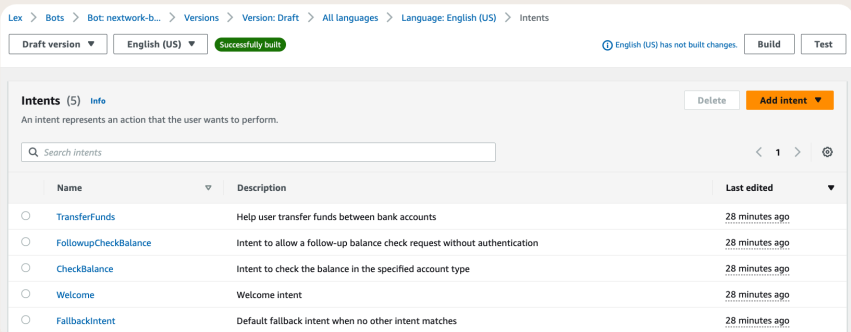
AWS CloudFormation

AWS CloudFormation is a service from Amazon Web Services that lets you create, configure, and deploy AWS resources automatically using templates, making it easy to set up complex infrastructure quickly, consistently, and without manual work.

I used CloudFormation to deploy BankerBot and its related AWS resources automatically in seconds, ensuring the chatbot was set up consistently and could be connected and tested without manually creating each component.



The final result!

Re-building my bot with CloudFormation took me just a few seconds, compared to the much longer time it would take to set everything up manually.

There was an error after I deployed my bot! The error was invalid bot configuration:access denied. I fixed this by creating and deploying a NEW Lambda function and finding the permission tab and Under the edit policy statement panel, select AWS service and entering the following details: Service: Other Statement ID: my-custom-permission-amazonlexchatbot Principal: lexv2.amazonaws.com Source name: find the ARN from the error message (eg. arn:aws:lex:ap-southeast-2:640168412593:bot-alias/*). Action: lamdaInvokeFunction

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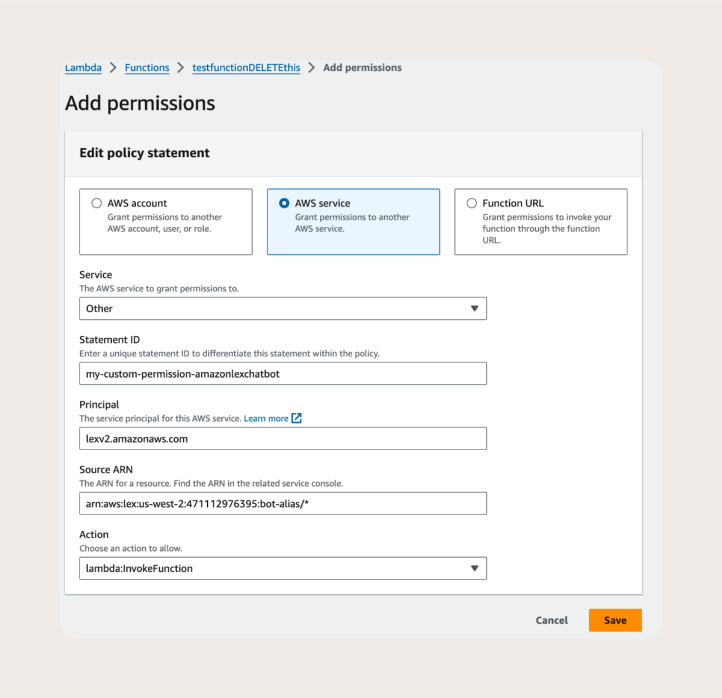
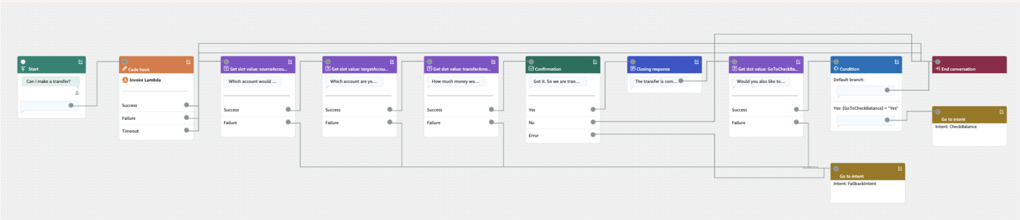


Using the visual builder

Using the visual builder, I added a new Get slot value card, a Condition card, and a Go to intent card after the transfer is completed. What this means for an end user is that after successfully transferring money, BankerBot will ask if they want to check an account balance and, if they say yes, seamlessly take them to the CheckBalance flow without restarting the conversation.

The new Get slot value card will trigger after the transfer is completed and the closing response is shown to the user. This card will try to capture a yes or no response (using the AMAZON.Confirmation slot type) to determine whether the user wants to check an account balance next.



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